

ASSESSMENT - CO1 AT3

NAME : PRENA M

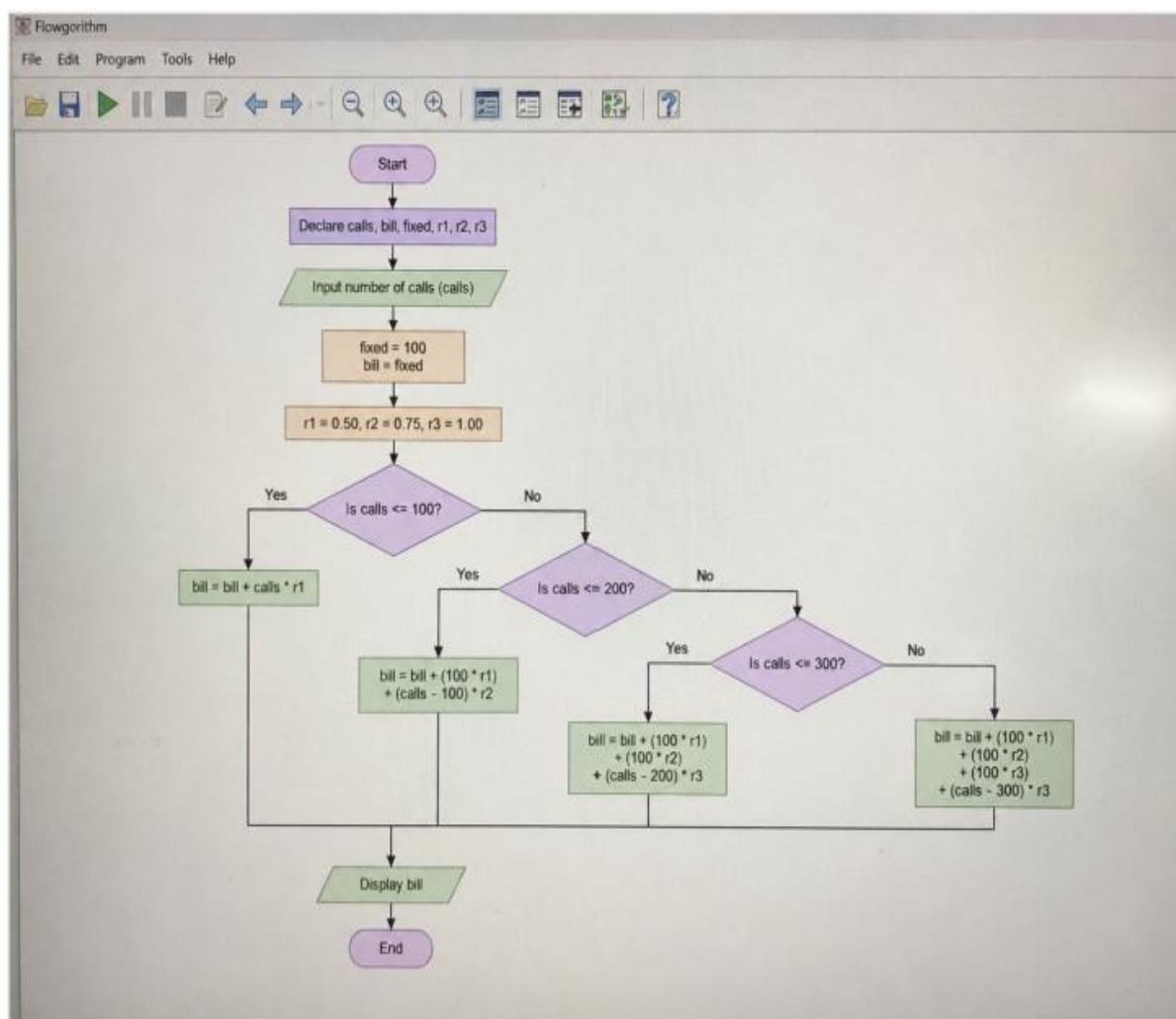
REG NO : 192524056

CODE : CSA 0924

COURSE : JAVA PROGRAMMING

1. Read the number of calls made and calculate a telephone bill with a fixed charge plus three usage slabs.

Flow chart:



Pseudo Code:

BEGIN

 DECLARE calls, bill

 DECLARE fixed = 100

 DECLARE rate1 = 0.50

 DECLARE rate2 = 0.75

 DECLARE rate3 = 1.00

 INPUT calls

 bill = fixed

 IF calls <= 100 THEN

 bill = bill + (calls * rate1)

 ELSE IF calls <= 200 THEN

 bill = bill + (100 * rate1)

 bill = bill + ((calls - 100) * rate2)

 ELSE

 bill = bill + (100 * rate1)

 bill = bill + (100 * rate2)

 bill = bill + ((calls - 200) * rate3)

 END IF

 OUTPUT "Telephone Bill = ", bill

END

Java program:

```
import java.util.Scanner;

public class TelephoneBill {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int calls;

        double bill;

        double fixed = 100;

        double rate1 = 0.50;

        double rate2 = 0.75;

        double rate3 = 1.00;

        System.out.print("Enter the number of calls: ");

        calls = sc.nextInt();

        bill = fixed;

        if (calls <= 100) {

            bill = bill + (calls * rate1);

        } else if (calls <= 200) {

            bill = bill + (100 * rate1)

                + ((calls - 100) * rate2);

        } else {

            bill = bill + (100 * rate1)

                + (100 * rate2)
```

```
        + ((calls - 200) * rate3);  
    }  
    System.out.println("Telephone Bill = " + bill);  
    sc.close();  
}  
}
```

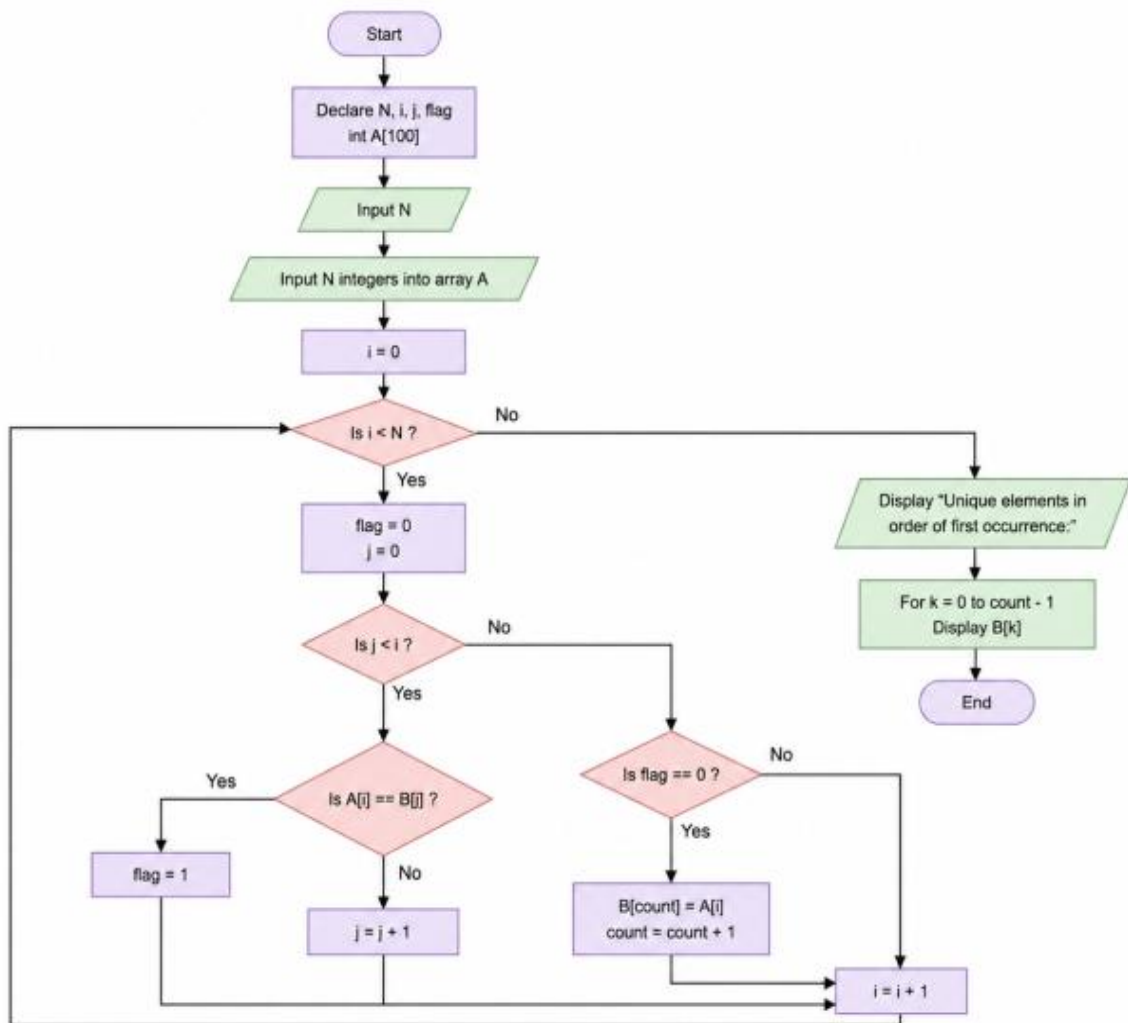
Sample Input:

Enter the number of calls: 250

Sample Output

Telephone Bill = 275.0

2. Read N integers and remove duplicate values while displaying only the first occurrence of each value.



Pseudo Code:

BEGIN

Declare N, i, j

Declare Array A[100]

Declare flag

INPUT N

FOR i ← 0 TO N-1

INPUT A[i]

```

END FOR

OUTPUT "Unique Elements:"

FOR i ← 0 TO N-1

    flag ← 0

    FOR j ← 0 TO i-1

        IF A[i] = A[j] THEN

            flag ← 1

            BREAK

        END IF

    END FOR

    IF flag = 0 THEN

        OUTPUT A[i]

    END IF

END FOR

END

```

Java program:

```

import java.util.Scanner;

public class RemoveDuplicates {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n;

        System.out.print("Enter the number of elements: ");

        n = sc.nextInt();

        int[] a = new int[n];
    }
}

```

```

        System.out.println("Enter the elements:");

        for (int i = 0; i < n; i++) {

            a[i] = sc.nextInt();

        }

        System.out.println("Unique Elements:");

        for (int i = 0; i < n; i++) {

            boolean flag = false;

            for (int j = 0; j < i; j++) {

                if (a[i] == a[j]) {

                    flag = true;

                    break;

                }

            }

            if (!flag) {

                System.out.print(a[i] + " ");

            }

        }

        sc.close();

    }

}

```

Sample Input:

Enter the number of elements: 7

Enter the elements:

10 20 10 30 20 40 30

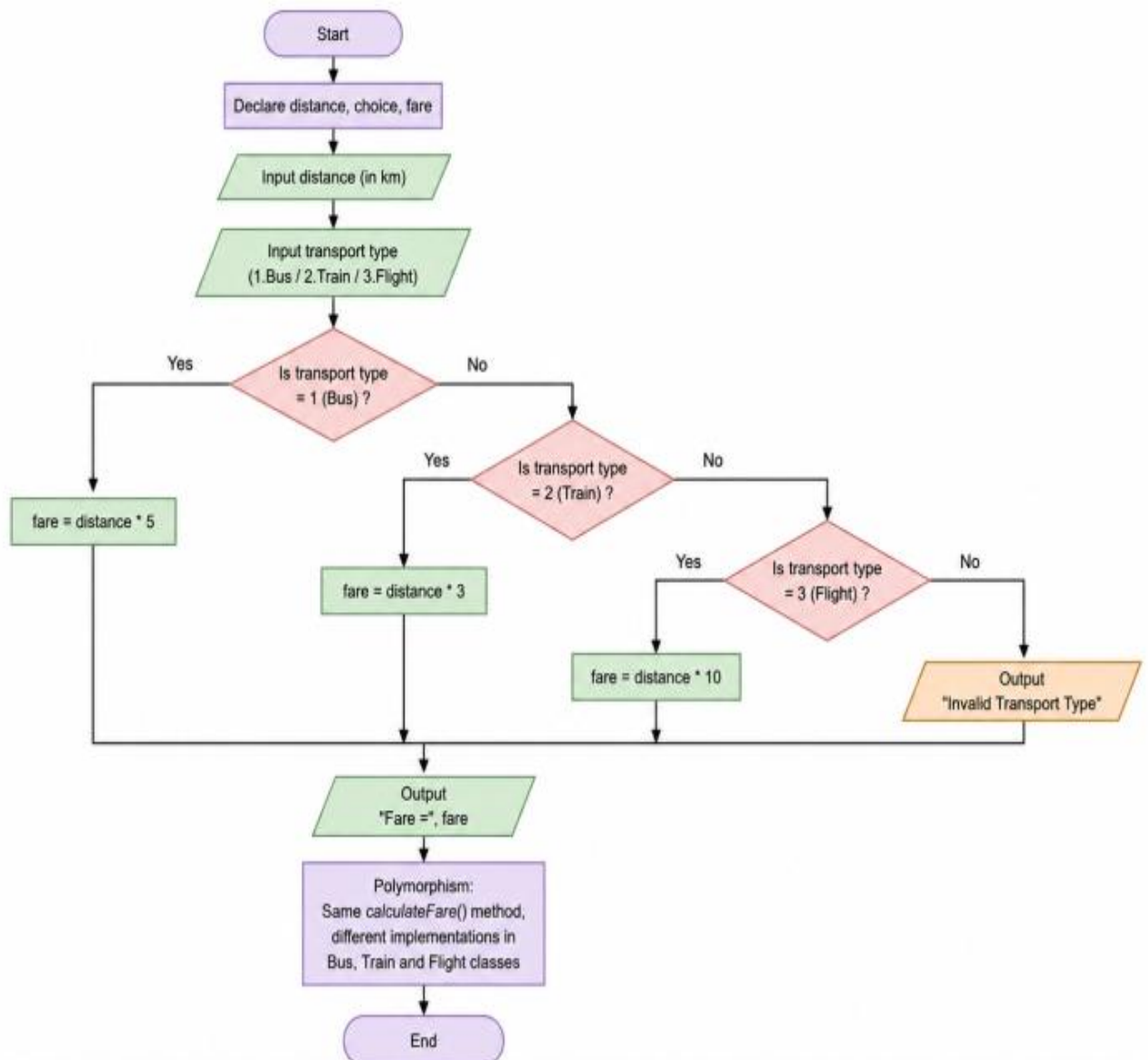
Sample Output:

Unique Elements:

10 20 30 40

3. Design a transport-fare flowchart for Bus, Train, and Flight. Use a common input process and different fare calculations, then relate it to polymorphism.

Flow chart:



Pseudo Code:

BEGIN

DECLARE distance, fare

DECLARE choice

OUTPUT "Transport Fare Calculation"

OUTPUT "1. Bus"

OUTPUT "2. Train"

```

OUTPUT "3. Flight"

INPUT distance

INPUT choice

IF choice = 1 THEN

    fare ← distance × 5

    OUTPUT "Transport : Bus"

ELSE IF choice = 2 THEN

    fare ← distance × 3

    OUTPUT "Transport : Train"

ELSE IF choice = 3 THEN

    fare ← distance × 10

    OUTPUT "Transport : Flight"

ELSE

    OUTPUT "Invalid Transport Type"

    STOP

END IF

OUTPUT "Distance = ", distance

OUTPUT "Fare = ", fare

OUTPUT "Polymorphism:"

OUTPUT "Same method (calculateFare)"

OUTPUT "Different fare calculation"

OUTPUT "for Bus, Train and Flight."

END

```

Java code:

```

import java.util.Scanner;

class TransportFare {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int distance, choice, fare = 0;

        System.out.print("Enter Distance: ");
    }
}

```

```
distance = sc.nextInt();

System.out.println("1.Bus 2.Train 3.Flight");

System.out.print("Enter Choice: ");

choice = sc.nextInt();

switch(choice) {

    case 1:

        fare = distance * 5;

        System.out.println("Transport: Bus");

        break;

    case 2:

        fare = distance * 3;

        System.out.println("Transport: Train");

        break;

    case 3:

        fare = distance * 10;

        System.out.println("Transport: Flight");

        break;

    default:

        System.out.println("Invalid Choice");

        return;

}

System.out.println("Fare = " + fare);

}
```

Input:

Enter Distance: 100

1. Bus
2. Train
3. Flight

Output:

Transport: Train

Fare = 300